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In [3]: import os
import pickle
import numpy as np
import tensorflow as tf
import matplotlib.pyplot as plt
from tensorflow.keras.preprocessing.image import ImageDataGenerator

# Dataset paths
train_dir = "/kaggle/input/datasets/sourajkundu/brain-tumor-dataset/brain tumor dataset/BT-MRI Dataset/BT-MRI Dataset/Training"
validation_dir = "/kaggle/input/datasets/sourajkundu/brain-tumor-dataset/brain tumor dataset/BT-MRI Dataset/BT-MRI Dataset/Validation"
test_dir = "/kaggle/input/datasets/sourajkundu/brain-tumor-dataset/brain tumor dataset/BT-MRI Dataset/BT-MRI Dataset/Testing"

# Parameters
img_height, img_width = 224, 224
batch_size = 100
epochs = 100
checkpoint_path = "/kaggle/working/VGG19_checkpoint.keras"
history_path = "/kaggle/working/VGG19_history.pkl"

class_labels = ['Glioma', 'Meningioma', 'No-tumor', 'Pituitary',]

# Data Generators
train_datagen = ImageDataGenerator(rescale=1.0 / 255)
validation_datagen = ImageDataGenerator(rescale=1.0 / 255)
test_datagen = ImageDataGenerator(rescale=1.0 / 255)

train_generator = train_datagen.flow_from_directory(
    train_dir, target_size=(img_height, img_width),
    batch_size=batch_size, class_mode='categorical'
)
validation_generator = validation_datagen.flow_from_directory(
    validation_dir, target_size=(img_height, img_width),
    batch_size=batch_size, class_mode='categorical'
)
test_generator = test_datagen.flow_from_directory(
    test_dir, target_size=(img_height, img_width),
    batch_size=batch_size, class_mode='categorical', shuffle=False
)

print(f"Number of images used for training: {train_generator.samples}")
print(f"Number of images used for validation: {validation_generator.samples}")
print(f"Number of images used for testing: {test_generator.samples}")

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Found 4512 images belonging to 4 classes.
Found 1600 images belonging to 4 classes.
Found 911 images belonging to 4 classes.
Number of images used for training: 4512
Number of images used for validation: 1600
Number of images used for testing: 911

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In [4]: from tensorflow.keras.models import Sequential
        from tensorflow.keras.layers import Flatten, Dense
        from tensorflow.keras.applications import VGG19

        # Build model
        model = Sequential()
        pretrained_model = VGG19(weights=None, include_top=False, input_shape=
(224, 224, 3))
        for layer in pretrained_model.layers:
            layer.trainable = False
        model.add(pretrained_model)
        model.add(Flatten())
        model.add(Dense(4096, activation='relu'))
        model.add(Dense(4096, activation='relu'))
        model.add(Dense(train_generator.num_classes, activation='softmax'))

        model.compile(
            optimizer='Adam',
            loss='categorical_crossentropy',
            metrics=[
                tf.keras.metrics.CategoricalAccuracy(name='accuracy'),
                tf.keras.metrics.Precision(name='precision'),
                tf.keras.metrics.Recall(name='recall'),
                tf.keras.metrics.AUC(name='auc')
            ]
        )

        # Load previous history if it exists (for resumable training)
        if os.path.exists(history_path):
            with open(history_path, "rb") as f:
                history = pickle.load(f)
        else:
            history = {k: [] for k in
                ["accuracy", "val_accuracy", "loss", "val_loss",
                "precision", "val_precision", "recall", "val_recall",
                "auc", "val_auc"]}

        checkpoint_callback = tf.keras.callbacks.ModelCheckpoint(
            filepath=checkpoint_path, save_best_only=False,
            save_weights_only=False, verbose=1
        )

        class SaveHistoryCallback(tf.keras.callbacks.Callback):
            def on_epoch_end(self, epoch, logs=None):
                for key in history:
                    history[key].append(logs[key])
                with open(history_path, "wb") as f:
                    pickle.dump(history, f)

        history_callback = SaveHistoryCallback()

        # Resume from checkpoint if it exists
        initial_epoch = 0
        if os.path.exists(checkpoint_path):
            model.load_weights(checkpoint_path)
            initial_epoch = len(history["accuracy"])
            print(f"Resuming training from epoch {initial_epoch}.")

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# Train
model.fit(
    train_generator,
    epochs=epochs,
    validation_data=validation_generator,
    initial_epoch=initial_epoch,
    callbacks=[checkpoint_callback, history_callback]
)

# Evaluate
results = model.evaluate(test_generator)
print("\nTest Results:")
print(f"Loss: {results[0]:.4f}, Accuracy: {results[1]:.4f}, Precision: {results[2]:.4f}, "
      f"Recall: {results[3]:.4f}, AUC: {results[4]:.4f}")
```

Epoch 1/100

1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m 632ms/step - accuracy: 0.3436 - auc: 0.6253 - loss: 1.2914 - precision: 0.7098 - recall: 0.0616
Epoch 1: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 1: finished saving model to /kaggle/working/VGG19_checkpoint.keras

1m46/46 [0m 32m] [0m 37m] [0m 1m51s] [0m 1s/step - accuracy: 0.4605 - auc: 0.7407 - loss: 1.1572 - precision: 0.8863 - recall: 0.1589 - val_accuracy: 0.6244 - val_auc: 0.8474 - val_loss: 1.0037 - val_precision: 0.7313 - val_recall: 0.3794

Epoch 2/100

1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m 600ms/step - accuracy: 0.6806 - auc: 0.8873 - loss: 0.8210 - precision: 0.7782 - recall: 0.5379
Epoch 2: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 2: finished saving model to /kaggle/working/VGG19_checkpoint.keras

1m46/46 [0m 32m] [0m 37m] [0m 1m45s] [0m 970ms/step - accuracy: 0.6753 - auc: 0.8905 - loss: 0.8007 - precision: 0.7551 - recall: 0.5612 - val_accuracy: 0.6988 - val_auc: 0.8976 - val_loss: 0.7829 - val_precision: 0.7648 - val_recall: 0.5956

Epoch 3/100

1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m 604ms/step - accuracy: 0.7235 - auc: 0.9164 - loss: 0.6988 - precision: 0.7760 - recall: 0.6404
Epoch 3: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 3: finished saving model to /kaggle/working/VGG19_checkpoint.keras

1m46/46 [0m 32m] [0m 37m] [0m 1m45s] [0m 977ms/step - accuracy: 0.7270 - auc: 0.9187 - loss: 0.6857 - precision: 0.7722 - recall: 0.6551 - val_accuracy: 0.7269 - val_auc: 0.9199 - val_loss: 0.6769 - val_precision: 0.7758 - val_recall: 0.6725

Epoch 4/100

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[1m46/46][0m [32m-----[0m[0m[37m[0m [1m0s[0m
599ms/step - accuracy: 0.7453 - auc: 0.9305 - loss: 0.6298 -
precision: 0.7801 - recall: 0.6854
Epoch 4: saving model to /kaggle/working/VGG19_checkpoint.keras

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Epoch 4: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46][0m [32m-----[0m[0m[37m[0m [1m45s[0m
973ms/step - accuracy: 0.7487 - auc: 0.9313 - loss: 0.6269 -
precision: 0.7848 - recall: 0.6944 - val_accuracy: 0.7581 -
val_auc: 0.9310 - val_loss: 0.6276 - val_precision: 0.7904 -
val_recall: 0.6931
Epoch 5/100
[1m46/46][0m [32m-----[0m[0m[37m[0m [1m0s[0m
602ms/step - accuracy: 0.7654 - auc: 0.9383 - loss: 0.5933 -
precision: 0.7941 - recall: 0.7073
Epoch 5: saving model to /kaggle/working/VGG19_checkpoint.keras

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Epoch 5: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46][0m [32m-----[0m[0m[37m[0m [1m45s[0m
977ms/step - accuracy: 0.7637 - auc: 0.9391 - loss: 0.5880 -
precision: 0.7932 - recall: 0.7117 - val_accuracy: 0.7150 -
val_auc: 0.9248 - val_loss: 0.6625 - val_precision: 0.7498 -
val_recall: 0.6800
Epoch 6/100
[1m46/46][0m [32m-----[0m[0m[37m[0m [1m0s[0m
602ms/step - accuracy: 0.7393 - auc: 0.9314 - loss: 0.6248 -
precision: 0.7896 - recall: 0.7031
Epoch 6: saving model to /kaggle/working/VGG19_checkpoint.keras

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Epoch 6: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46][0m [32m-----[0m[0m[37m[0m [1m45s[0m
975ms/step - accuracy: 0.7555 - auc: 0.9384 - loss: 0.5887 -
precision: 0.7978 - recall: 0.7154 - val_accuracy: 0.7775 -
val_auc: 0.9419 - val_loss: 0.5697 - val_precision: 0.8058 -
val_recall: 0.7469
Epoch 7/100
[1m46/46][0m [32m-----[0m[0m[37m[0m [1m0s[0m
599ms/step - accuracy: 0.7969 - auc: 0.9514 - loss: 0.5212 -
precision: 0.8169 - recall: 0.7659
Epoch 7: saving model to /kaggle/working/VGG19_checkpoint.keras

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Epoch 7: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46][0m [32m-----[0m[0m[37m[0m [1m45s[0m
971ms/step - accuracy: 0.8045 - auc: 0.9548 - loss: 0.5028 -
precision: 0.8247 - recall: 0.7735 - val_accuracy: 0.7688 -
val_auc: 0.9479 - val_loss: 0.5383 - val_precision: 0.7865 -
val_recall: 0.7412
Epoch 8/100
[1m46/46][0m [32m-----[0m[0m[37m[0m [1m0s[0m
604ms/step - accuracy: 0.8202 - auc: 0.9593 - loss: 0.4734 -
precision: 0.8378 - recall: 0.7917
Epoch 8: saving model to /kaggle/working/VGG19_checkpoint.keras

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Epoch 8: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m [0m[37m[0m [1m45s[0m
977ms/step - accuracy: 0.8229 - auc: 0.9609 - loss: 0.4666 -
precision: 0.8425 - recall: 0.7952 - val_accuracy: 0.8069 -
val_auc: 0.9584 - val_loss: 0.4789 - val_precision: 0.8274 -
val_recall: 0.7819
Epoch 9/100
[1m46/46[0m [32m-----[0m [0m[37m[0m [1m0s[0m
600ms/step - accuracy: 0.8250 - auc: 0.9615 - loss: 0.4618 -
precision: 0.8401 - recall: 0.8021
Epoch 9: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 9: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m [0m[37m[0m [1m45s[0m
973ms/step - accuracy: 0.8098 - auc: 0.9578 - loss: 0.4837 -
precision: 0.8282 - recall: 0.7872 - val_accuracy: 0.8156 -
val_auc: 0.9600 - val_loss: 0.4698 - val_precision: 0.8295 -
val_recall: 0.7969
Epoch 10/100
[1m46/46[0m [32m-----[0m [0m[37m[0m [1m0s[0m
602ms/step - accuracy: 0.8481 - auc: 0.9697 - loss: 0.4081 -
precision: 0.8605 - recall: 0.8330
Epoch 10: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 10: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m [0m[37m[0m [1m45s[0m
973ms/step - accuracy: 0.8411 - auc: 0.9684 - loss: 0.4161 -
precision: 0.8548 - recall: 0.8258 - val_accuracy: 0.8250 -
val_auc: 0.9591 - val_loss: 0.4704 - val_precision: 0.8397 -
val_recall: 0.8087
Epoch 11/100
[1m46/46[0m [32m-----[0m [0m[37m[0m [1m0s[0m
603ms/step - accuracy: 0.8455 - auc: 0.9690 - loss: 0.4112 -
precision: 0.8589 - recall: 0.8293
Epoch 11: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 11: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m [0m[37m[0m [1m45s[0m
971ms/step - accuracy: 0.8446 - auc: 0.9685 - loss: 0.4145 -
precision: 0.8591 - recall: 0.8298 - val_accuracy: 0.8006 -
val_auc: 0.9555 - val_loss: 0.4935 - val_precision: 0.8176 -
val_recall: 0.7819
Epoch 12/100
[1m46/46[0m [32m-----[0m [0m[37m[0m [1m0s[0m
600ms/step - accuracy: 0.8551 - auc: 0.9718 - loss: 0.3888 -
precision: 0.8676 - recall: 0.8382
Epoch 12: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 12: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m [0m[37m[0m [1m44s[0m
967ms/step - accuracy: 0.8568 - auc: 0.9732 - loss: 0.3813 -
precision: 0.8694 - recall: 0.8382 - val_accuracy: 0.8219 -

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val_auc: 0.9628 - val_loss: 0.4510 - val_precision: 0.8310 -
 val_recall: 0.8019
 Epoch 13/100
 1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m 603ms/step - accuracy: 0.8528 - auc: 0.9734 - loss: 0.3787 -
 precision: 0.8632 - recall: 0.8393
 Epoch 13: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 13: finished saving model to
 /kaggle/working/VGG19_checkpoint.keras
 1m46/46 [0m 32m] [0m 37m] [0m 1m45s] [0m 972ms/step - accuracy: 0.8375 - auc: 0.9688 - loss: 0.4124 -
 precision: 0.8500 - recall: 0.8236 - val_accuracy: 0.8194 -
 val_auc: 0.9600 - val_loss: 0.4686 - val_precision: 0.8316 -
 val_recall: 0.7931
 Epoch 14/100
 1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m 599ms/step - accuracy: 0.8650 - auc: 0.9749 - loss: 0.3666 -
 precision: 0.8775 - recall: 0.8514
 Epoch 14: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 14: finished saving model to
 /kaggle/working/VGG19_checkpoint.keras
 1m46/46 [0m 32m] [0m 37m] [0m 1m44s] [0m 965ms/step - accuracy: 0.8703 - auc: 0.9773 - loss: 0.3501 -
 precision: 0.8819 - recall: 0.8570 - val_accuracy: 0.8350 -
 val_auc: 0.9707 - val_loss: 0.3982 - val_precision: 0.8469 -
 val_recall: 0.8231
 Epoch 15/100
 1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m 604ms/step - accuracy: 0.8697 - auc: 0.9772 - loss: 0.3481 -
 precision: 0.8760 - recall: 0.8575
 Epoch 15: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 15: finished saving model to
 /kaggle/working/VGG19_checkpoint.keras
 1m46/46 [0m 32m] [0m 37m] [0m 1m45s] [0m 971ms/step - accuracy: 0.8646 - auc: 0.9770 - loss: 0.3497 -
 precision: 0.8732 - recall: 0.8544 - val_accuracy: 0.8537 -
 val_auc: 0.9744 - val_loss: 0.3709 - val_precision: 0.8635 -
 val_recall: 0.8419
 Epoch 16/100
 1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m 600ms/step - accuracy: 0.8976 - auc: 0.9827 - loss: 0.2992 -
 precision: 0.9042 - recall: 0.8875
 Epoch 16: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 16: finished saving model to
 /kaggle/working/VGG19_checkpoint.keras
 1m46/46 [0m 32m] [0m 37m] [0m 1m45s] [0m 969ms/step - accuracy: 0.8870 - auc: 0.9814 - loss: 0.3122 -
 precision: 0.8939 - recall: 0.8774 - val_accuracy: 0.8500 -
 val_auc: 0.9720 - val_loss: 0.3877 - val_precision: 0.8619 -
 val_recall: 0.8350
 Epoch 17/100
 1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m 601ms/step - accuracy: 0.8823 - auc: 0.9825 - loss: 0.3043 -

precision: 0.8901 - recall: 0.8733
 Epoch 17: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 17: finished saving model to
 /kaggle/working/VGG19_checkpoint.keras
 1m46/46[0m 32m-----[0m[37m[0m 1m45s[0m
 970ms/step - accuracy: 0.8887 - auc: 0.9833 - loss: 0.2959 -
 precision: 0.8951 - recall: 0.8797 - val_accuracy: 0.8506 -
 val_auc: 0.9740 - val_loss: 0.3738 - val_precision: 0.8592 -
 val_recall: 0.8431

Epoch 18/100
 1m46/46[0m 32m-----[0m[37m[0m 1m0s[0m
 602ms/step - accuracy: 0.8904 - auc: 0.9842 - loss: 0.2908 -
 precision: 0.8990 - recall: 0.8830
 Epoch 18: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 18: finished saving model to
 /kaggle/working/VGG19_checkpoint.keras
 1m46/46[0m 32m-----[0m[37m[0m 1m45s[0m
 970ms/step - accuracy: 0.8921 - auc: 0.9845 - loss: 0.2855 -
 precision: 0.9019 - recall: 0.8839 - val_accuracy: 0.8612 -
 val_auc: 0.9765 - val_loss: 0.3509 - val_precision: 0.8710 -
 val_recall: 0.8525

Epoch 19/100
 1m46/46[0m 32m-----[0m[37m[0m 1m0s[0m
 600ms/step - accuracy: 0.8857 - auc: 0.9817 - loss: 0.3073 -
 precision: 0.8943 - recall: 0.8762
 Epoch 19: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 19: finished saving model to
 /kaggle/working/VGG19_checkpoint.keras
 1m46/46[0m 32m-----[0m[37m[0m 1m45s[0m
 971ms/step - accuracy: 0.8823 - auc: 0.9821 - loss: 0.3066 -
 precision: 0.8900 - recall: 0.8730 - val_accuracy: 0.8788 -
 val_auc: 0.9799 - val_loss: 0.3263 - val_precision: 0.8802 -
 val_recall: 0.8675

Epoch 20/100
 1m46/46[0m 32m-----[0m[37m[0m 1m0s[0m
 601ms/step - accuracy: 0.9020 - auc: 0.9861 - loss: 0.2709 -
 precision: 0.9091 - recall: 0.8958
 Epoch 20: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 20: finished saving model to
 /kaggle/working/VGG19_checkpoint.keras
 1m46/46[0m 32m-----[0m[37m[0m 1m44s[0m
 967ms/step - accuracy: 0.9016 - auc: 0.9860 - loss: 0.2703 -
 precision: 0.9084 - recall: 0.8949 - val_accuracy: 0.8669 -
 val_auc: 0.9751 - val_loss: 0.3609 - val_precision: 0.8755 -
 val_recall: 0.8569

Epoch 21/100
 1m46/46[0m 32m-----[0m[37m[0m 1m0s[0m
 602ms/step - accuracy: 0.9099 - auc: 0.9887 - loss: 0.2431 -
 precision: 0.9143 - recall: 0.9041
 Epoch 21: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 21: finished saving model to
 /kaggle/working/VGG19_checkpoint.keras

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[1m46/46][0m [32m-----[0m[0m[37m[0m [1m45s[0m
971ms/step - accuracy: 0.9038 - auc: 0.9876 - loss: 0.2549 -
precision: 0.9107 - recall: 0.8969 - val_accuracy: 0.8644 -
val_auc: 0.9773 - val_loss: 0.3490 - val_precision: 0.8717 -
val_recall: 0.8581

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Epoch 22/100

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[1m46/46][0m [32m-----[0m[0m[37m[0m [1m0s[0m
600ms/step - accuracy: 0.9040 - auc: 0.9881 - loss: 0.2496 -
precision: 0.9094 - recall: 0.8959

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Epoch 22: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 22: finished saving model to
/kaggle/working/VGG19_checkpoint.keras

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[1m46/46][0m [32m-----[0m[0m[37m[0m [1m44s[0m
967ms/step - accuracy: 0.9062 - auc: 0.9879 - loss: 0.2485 -
precision: 0.9124 - recall: 0.9000 - val_accuracy: 0.8756 -
val_auc: 0.9796 - val_loss: 0.3282 - val_precision: 0.8863 -
val_recall: 0.8675

```

Epoch 23/100

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[1m46/46][0m [32m-----[0m[0m[37m[0m [1m0s[0m
604ms/step - accuracy: 0.9065 - auc: 0.9880 - loss: 0.2491 -
precision: 0.9138 - recall: 0.9023

```

Epoch 23: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 23: finished saving model to
/kaggle/working/VGG19_checkpoint.keras

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[1m46/46][0m [32m-----[0m[0m[37m[0m [1m45s[0m
971ms/step - accuracy: 0.9051 - auc: 0.9880 - loss: 0.2478 -
precision: 0.9118 - recall: 0.9009 - val_accuracy: 0.8813 -
val_auc: 0.9803 - val_loss: 0.3188 - val_precision: 0.8887 -
val_recall: 0.8737

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Epoch 24/100

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[1m46/46][0m [32m-----[0m[0m[37m[0m [1m0s[0m
600ms/step - accuracy: 0.9202 - auc: 0.9903 - loss: 0.2226 -
precision: 0.9258 - recall: 0.9150

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Epoch 24: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 24: finished saving model to
/kaggle/working/VGG19_checkpoint.keras

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[1m46/46][0m [32m-----[0m[0m[37m[0m [1m44s[0m
965ms/step - accuracy: 0.9127 - auc: 0.9892 - loss: 0.2336 -
precision: 0.9173 - recall: 0.9076 - val_accuracy: 0.8612 -
val_auc: 0.9777 - val_loss: 0.3489 - val_precision: 0.8682 -
val_recall: 0.8562

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Epoch 25/100

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[1m46/46][0m [32m-----[0m[0m[37m[0m [1m0s[0m
603ms/step - accuracy: 0.8849 - auc: 0.9822 - loss: 0.3040 -
precision: 0.8898 - recall: 0.8759

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Epoch 25: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 25: finished saving model to
/kaggle/working/VGG19_checkpoint.keras

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[1m46/46][0m [32m-----[0m[0m[37m[0m [1m45s[0m
969ms/step - accuracy: 0.8958 - auc: 0.9842 - loss: 0.2836 -
precision: 0.9029 - recall: 0.8881 - val_accuracy: 0.8931 -
val_auc: 0.9847 - val_loss: 0.2850 - val_precision: 0.8980 -
val_recall: 0.8856

```


Epoch 26/100

1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m 598ms/step - accuracy: 0.9113 - auc: 0.9895 - loss: 0.2301 - precision: 0.9176 - recall: 0.9068
Epoch 26: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 26: finished saving model to

/kaggle/working/VGG19_checkpoint.keras

1m46/46 [0m 32m] [0m 37m] [0m 1m44s] [0m 964ms/step - accuracy: 0.9151 - auc: 0.9902 - loss: 0.2233 - precision: 0.9202 - recall: 0.9102 - val_accuracy: 0.8869 - val_auc: 0.9826 - val_loss: 0.3004 - val_precision: 0.8889 - val_recall: 0.8800

Epoch 27/100

1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m 602ms/step - accuracy: 0.9030 - auc: 0.9879 - loss: 0.2510 - precision: 0.9080 - recall: 0.8967

Epoch 27: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 27: finished saving model to

/kaggle/working/VGG19_checkpoint.keras

1m46/46 [0m 32m] [0m 37m] [0m 1m44s] [0m 968ms/step - accuracy: 0.9120 - auc: 0.9892 - loss: 0.2344 - precision: 0.9162 - recall: 0.9060 - val_accuracy: 0.7987 - val_auc: 0.9527 - val_loss: 0.5412 - val_precision: 0.8172 - val_recall: 0.7850

Epoch 28/100

1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m 602ms/step - accuracy: 0.8674 - auc: 0.9805 - loss: 0.3238 - precision: 0.8757 - recall: 0.8565

Epoch 28: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 28: finished saving model to

/kaggle/working/VGG19_checkpoint.keras

1m46/46 [0m 32m] [0m 37m] [0m 1m45s] [0m 972ms/step - accuracy: 0.8839 - auc: 0.9839 - loss: 0.2903 - precision: 0.8907 - recall: 0.8741 - val_accuracy: 0.8875 - val_auc: 0.9826 - val_loss: 0.3020 - val_precision: 0.8913 - val_recall: 0.8813

Epoch 29/100

1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m 602ms/step - accuracy: 0.9305 - auc: 0.9943 - loss: 0.1764 - precision: 0.9380 - recall: 0.9280

Epoch 29: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 29: finished saving model to

/kaggle/working/VGG19_checkpoint.keras

1m46/46 [0m 32m] [0m 37m] [0m 1m45s] [0m 973ms/step - accuracy: 0.9313 - auc: 0.9938 - loss: 0.1801 - precision: 0.9365 - recall: 0.9284 - val_accuracy: 0.8744 - val_auc: 0.9820 - val_loss: 0.3154 - val_precision: 0.8785 - val_recall: 0.8675

Epoch 30/100

1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m 602ms/step - accuracy: 0.9300 - auc: 0.9933 - loss: 0.1830 - precision: 0.9337 - recall: 0.9260

Epoch 30: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 30: finished saving model to
 /kaggle/working/VGG19_checkpoint.keras
 1m46/46[0m 32m[0m 37m[0m 1m45s[0m
 972ms/step - accuracy: 0.9284 - auc: 0.9926 - loss: 0.1904 -
 precision: 0.9314 - recall: 0.9238 - val_accuracy: 0.8731 -
 val_auc: 0.9809 - val_loss: 0.3241 - val_precision: 0.8770 -
 val_recall: 0.8694

Epoch 31/100
 1m46/46[0m 32m[0m 37m[0m 1m0s[0m
 602ms/step - accuracy: 0.9257 - auc: 0.9930 - loss: 0.1919 -
 precision: 0.9272 - recall: 0.9233
 Epoch 31: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 31: finished saving model to
 /kaggle/working/VGG19_checkpoint.keras
 1m46/46[0m 32m[0m 37m[0m 1m45s[0m
 968ms/step - accuracy: 0.9295 - auc: 0.9931 - loss: 0.1865 -
 precision: 0.9324 - recall: 0.9266 - val_accuracy: 0.9013 -
 val_auc: 0.9856 - val_loss: 0.2723 - val_precision: 0.9055 -
 val_recall: 0.8981

Epoch 32/100
 1m46/46[0m 32m[0m 37m[0m 1m0s[0m
 603ms/step - accuracy: 0.9455 - auc: 0.9953 - loss: 0.1539 -
 precision: 0.9488 - recall: 0.9430
 Epoch 32: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 32: finished saving model to
 /kaggle/working/VGG19_checkpoint.keras
 1m46/46[0m 32m[0m 37m[0m 1m45s[0m
 972ms/step - accuracy: 0.9461 - auc: 0.9958 - loss: 0.1480 -
 precision: 0.9492 - recall: 0.9439 - val_accuracy: 0.8981 -
 val_auc: 0.9860 - val_loss: 0.2678 - val_precision: 0.9029 -
 val_recall: 0.8950

Epoch 33/100
 1m46/46[0m 32m[0m 37m[0m 1m0s[0m
 601ms/step - accuracy: 0.9591 - auc: 0.9968 - loss: 0.1277 -
 precision: 0.9600 - recall: 0.9575
 Epoch 33: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 33: finished saving model to
 /kaggle/working/VGG19_checkpoint.keras
 1m46/46[0m 32m[0m 37m[0m 1m45s[0m
 969ms/step - accuracy: 0.9599 - auc: 0.9968 - loss: 0.1255 -
 precision: 0.9613 - recall: 0.9579 - val_accuracy: 0.9100 -
 val_auc: 0.9899 - val_loss: 0.2263 - val_precision: 0.9166 -
 val_recall: 0.9069

Epoch 34/100
 1m46/46[0m 32m[0m 37m[0m 1m0s[0m
 602ms/step - accuracy: 0.9545 - auc: 0.9970 - loss: 0.1261 -
 precision: 0.9576 - recall: 0.9515
 Epoch 34: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 34: finished saving model to
 /kaggle/working/VGG19_checkpoint.keras
 1m46/46[0m 32m[0m 37m[0m 1m45s[0m
 969ms/step - accuracy: 0.9550 - auc: 0.9970 - loss: 0.1227 -

```
precision: 0.9579 - recall: 0.9535 - val_accuracy: 0.8844 -
val_auc: 0.9807 - val_loss: 0.3238 - val_precision: 0.8870 -
val_recall: 0.8781
```

Epoch 35/100

```
1m46/46 [0m 32m-----[0m 37m [0m 1m0s [0m
601ms/step - accuracy: 0.9065 - auc: 0.9868 - loss: 0.2541 -
precision: 0.9124 - recall: 0.9022
```

Epoch 35: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 35: finished saving model to

/kaggle/working/VGG19_checkpoint.keras

```
1m46/46 [0m 32m-----[0m 37m [0m 1m44s [0m
966ms/step - accuracy: 0.8947 - auc: 0.9855 - loss: 0.2746 -
precision: 0.8997 - recall: 0.8885 - val_accuracy: 0.8881 -
val_auc: 0.9829 - val_loss: 0.2984 - val_precision: 0.8941 -
val_recall: 0.8813
```

Epoch 36/100

```
1m46/46 [0m 32m-----[0m 37m [0m 1m0s [0m
602ms/step - accuracy: 0.9485 - auc: 0.9962 - loss: 0.1448 -
precision: 0.9500 - recall: 0.9453
```

Epoch 36: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 36: finished saving model to

/kaggle/working/VGG19_checkpoint.keras

```
1m46/46 [0m 32m-----[0m 37m [0m 1m45s [0m
971ms/step - accuracy: 0.9515 - auc: 0.9964 - loss: 0.1403 -
precision: 0.9534 - recall: 0.9484 - val_accuracy: 0.9112 -
val_auc: 0.9888 - val_loss: 0.2378 - val_precision: 0.9150 -
val_recall: 0.9087
```

Epoch 37/100

```
1m46/46 [0m 32m-----[0m 37m [0m 1m0s [0m
599ms/step - accuracy: 0.9594 - auc: 0.9972 - loss: 0.1208 -
precision: 0.9606 - recall: 0.9575
```

Epoch 37: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 37: finished saving model to

/kaggle/working/VGG19_checkpoint.keras

```
1m46/46 [0m 32m-----[0m 37m [0m 1m44s [0m
965ms/step - accuracy: 0.9552 - auc: 0.9966 - loss: 0.1261 -
precision: 0.9569 - recall: 0.9535 - val_accuracy: 0.9137 -
val_auc: 0.9885 - val_loss: 0.2370 - val_precision: 0.9163 -
val_recall: 0.9100
```

Epoch 38/100

```
1m46/46 [0m 32m-----[0m 37m [0m 1m0s [0m
603ms/step - accuracy: 0.9698 - auc: 0.9981 - loss: 0.1001 -
precision: 0.9715 - recall: 0.9673
```

Epoch 38: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 38: finished saving model to

/kaggle/working/VGG19_checkpoint.keras

```
1m46/46 [0m 32m-----[0m 37m [0m 1m45s [0m
970ms/step - accuracy: 0.9703 - auc: 0.9980 - loss: 0.0990 -
precision: 0.9717 - recall: 0.9681 - val_accuracy: 0.9169 -
val_auc: 0.9905 - val_loss: 0.2213 - val_precision: 0.9218 -
val_recall: 0.9131
```

Epoch 39/100

```
1m46/46 [0m 32m-----[0m 37m [0m 1m0s [0m
```

600ms/step - accuracy: 0.9683 - auc: 0.9987 - loss: 0.0861 -
precision: 0.9696 - recall: 0.9677
Epoch 39: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 39: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
1m46/46[0m 32m-----[0m[37m[0m 1m44s[0m
964ms/step - accuracy: 0.9605 - auc: 0.9975 - loss: 0.1058 -
precision: 0.9620 - recall: 0.9592 - val_accuracy: 0.8213 -
val_auc: 0.9611 - val_loss: 0.5085 - val_precision: 0.8274 -
val_recall: 0.8150
Epoch 40/100
1m46/46[0m 32m-----[0m[37m[0m 1m0s[0m
602ms/step - accuracy: 0.9375 - auc: 0.9946 - loss: 0.1567 -
precision: 0.9387 - recall: 0.9364
Epoch 40: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 40: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
1m46/46[0m 32m-----[0m[37m[0m 1m44s[0m
967ms/step - accuracy: 0.9555 - auc: 0.9968 - loss: 0.1212 -
precision: 0.9571 - recall: 0.9546 - val_accuracy: 0.9237 -
val_auc: 0.9898 - val_loss: 0.2191 - val_precision: 0.9260 -
val_recall: 0.9231
Epoch 41/100
1m46/46[0m 32m-----[0m[37m[0m 1m0s[0m
601ms/step - accuracy: 0.9823 - auc: 0.9993 - loss: 0.0671 -
precision: 0.9824 - recall: 0.9816
Epoch 41: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 41: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
1m46/46[0m 32m-----[0m[37m[0m 1m45s[0m
970ms/step - accuracy: 0.9721 - auc: 0.9977 - loss: 0.0963 -
precision: 0.9729 - recall: 0.9716 - val_accuracy: 0.9106 -
val_auc: 0.9882 - val_loss: 0.2399 - val_precision: 0.9168 -
val_recall: 0.9087
Epoch 42/100
1m46/46[0m 32m-----[0m[37m[0m 1m0s[0m
601ms/step - accuracy: 0.9369 - auc: 0.9947 - loss: 0.1603 -
precision: 0.9388 - recall: 0.9341
Epoch 42: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 42: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
1m46/46[0m 32m-----[0m[37m[0m 1m45s[0m
968ms/step - accuracy: 0.9504 - auc: 0.9963 - loss: 0.1328 -
precision: 0.9517 - recall: 0.9481 - val_accuracy: 0.9244 -
val_auc: 0.9907 - val_loss: 0.2107 - val_precision: 0.9253 -
val_recall: 0.9206
Epoch 43/100
1m46/46[0m 32m-----[0m[37m[0m 1m0s[0m
601ms/step - accuracy: 0.9791 - auc: 0.9990 - loss: 0.0717 -
precision: 0.9797 - recall: 0.9790
Epoch 43: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 43: finished saving model to

```

/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m[37m[0m [1m44s[0m
968ms/step - accuracy: 0.9840 - auc: 0.9992 - loss: 0.0616 -
precision: 0.9847 - recall: 0.9838 - val_accuracy: 0.9331 -
val_auc: 0.9914 - val_loss: 0.1970 - val_precision: 0.9341 -
val_recall: 0.9300
Epoch 44/100
[1m46/46[0m [32m-----[0m[37m[0m [1m0s[0m
601ms/step - accuracy: 0.9819 - auc: 0.9994 - loss: 0.0601 -
precision: 0.9820 - recall: 0.9819
Epoch 44: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 44: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m[37m[0m [1m45s[0m
968ms/step - accuracy: 0.9774 - auc: 0.9992 - loss: 0.0695 -
precision: 0.9776 - recall: 0.9772 - val_accuracy: 0.9150 -
val_auc: 0.9883 - val_loss: 0.2461 - val_precision: 0.9166 -
val_recall: 0.9137
Epoch 45/100
[1m46/46[0m [32m-----[0m[37m[0m [1m0s[0m
602ms/step - accuracy: 0.9785 - auc: 0.9992 - loss: 0.0670 -
precision: 0.9788 - recall: 0.9785
Epoch 45: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 45: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m[37m[0m [1m45s[0m
970ms/step - accuracy: 0.9807 - auc: 0.9993 - loss: 0.0628 -
precision: 0.9812 - recall: 0.9807 - val_accuracy: 0.9262 -
val_auc: 0.9913 - val_loss: 0.2031 - val_precision: 0.9308 -
val_recall: 0.9250
Epoch 46/100
[1m46/46[0m [32m-----[0m[37m[0m [1m0s[0m
597ms/step - accuracy: 0.9783 - auc: 0.9992 - loss: 0.0649 -
precision: 0.9784 - recall: 0.9782
Epoch 46: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 46: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m[37m[0m [1m44s[0m
965ms/step - accuracy: 0.9792 - auc: 0.9991 - loss: 0.0640 -
precision: 0.9796 - recall: 0.9787 - val_accuracy: 0.7431 -
val_auc: 0.9207 - val_loss: 0.9396 - val_precision: 0.7436 -
val_recall: 0.7431
Epoch 47/100
[1m46/46[0m [32m-----[0m[37m[0m [1m0s[0m
602ms/step - accuracy: 0.8249 - auc: 0.9600 - loss: 0.5530 -
precision: 0.8314 - recall: 0.8218
Epoch 47: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 47: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m[37m[0m [1m45s[0m
969ms/step - accuracy: 0.8542 - auc: 0.9707 - loss: 0.4220 -
precision: 0.8620 - recall: 0.8497 - val_accuracy: 0.9038 -
val_auc: 0.9866 - val_loss: 0.2611 - val_precision: 0.9098 -

```

```

val_recall: 0.9013
Epoch 48/100
[1m46/46[0m [32m-----[0m[37m[0m [1m0s[0m
600ms/step - accuracy: 0.9550 - auc: 0.9965 - loss: 0.1376 -
precision: 0.9581 - recall: 0.9511
Epoch 48: saving model to /kaggle/working/VGG19_checkpoint.keras

```

```

Epoch 48: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m[37m[0m [1m44s[0m
967ms/step - accuracy: 0.9566 - auc: 0.9971 - loss: 0.1296 -
precision: 0.9596 - recall: 0.9526 - val_accuracy: 0.8712 -
val_auc: 0.9769 - val_loss: 0.3621 - val_precision: 0.8732 -
val_recall: 0.8650
Epoch 49/100
[1m46/46[0m [32m-----[0m[37m[0m [1m0s[0m
603ms/step - accuracy: 0.9597 - auc: 0.9975 - loss: 0.1164 -
precision: 0.9619 - recall: 0.9584
Epoch 49: saving model to /kaggle/working/VGG19_checkpoint.keras

```

```

Epoch 49: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m[37m[0m [1m44s[0m
968ms/step - accuracy: 0.9643 - auc: 0.9979 - loss: 0.1092 -
precision: 0.9657 - recall: 0.9623 - val_accuracy: 0.9300 -
val_auc: 0.9912 - val_loss: 0.2016 - val_precision: 0.9309 -
val_recall: 0.9256
Epoch 50/100
[1m46/46[0m [32m-----[0m[37m[0m [1m0s[0m
600ms/step - accuracy: 0.9830 - auc: 0.9987 - loss: 0.0762 -
precision: 0.9833 - recall: 0.9825
Epoch 50: saving model to /kaggle/working/VGG19_checkpoint.keras

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```

Epoch 50: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m[37m[0m [1m44s[0m
967ms/step - accuracy: 0.9767 - auc: 0.9989 - loss: 0.0814 -
precision: 0.9776 - recall: 0.9758 - val_accuracy: 0.9275 -
val_auc: 0.9910 - val_loss: 0.2044 - val_precision: 0.9313 -
val_recall: 0.9237
Epoch 51/100
[1m46/46[0m [32m-----[0m[37m[0m [1m0s[0m
602ms/step - accuracy: 0.9742 - auc: 0.9987 - loss: 0.0884 -
precision: 0.9751 - recall: 0.9726
Epoch 51: saving model to /kaggle/working/VGG19_checkpoint.keras

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```

Epoch 51: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m[37m[0m [1m44s[0m
967ms/step - accuracy: 0.9738 - auc: 0.9988 - loss: 0.0853 -
precision: 0.9744 - recall: 0.9714 - val_accuracy: 0.8644 -
val_auc: 0.9810 - val_loss: 0.3570 - val_precision: 0.8706 -
val_recall: 0.8619
Epoch 52/100
[1m46/46[0m [32m-----[0m[37m[0m [1m0s[0m
601ms/step - accuracy: 0.9523 - auc: 0.9964 - loss: 0.1308 -
precision: 0.9548 - recall: 0.9487

```

```

Epoch 52: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m[37m[0m [1m44s[0m
967ms/step - accuracy: 0.9738 - auc: 0.9988 - loss: 0.0853 -
precision: 0.9744 - recall: 0.9714 - val_accuracy: 0.8644 -
val_auc: 0.9810 - val_loss: 0.3570 - val_precision: 0.8706 -
val_recall: 0.8619
Epoch 52/100
[1m46/46[0m [32m-----[0m[37m[0m [1m0s[0m
601ms/step - accuracy: 0.9523 - auc: 0.9964 - loss: 0.1308 -
precision: 0.9548 - recall: 0.9487

```

```

Epoch 52: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m[37m[0m [1m44s[0m
967ms/step - accuracy: 0.9738 - auc: 0.9988 - loss: 0.0853 -
precision: 0.9744 - recall: 0.9714 - val_accuracy: 0.8644 -
val_auc: 0.9810 - val_loss: 0.3570 - val_precision: 0.8706 -
val_recall: 0.8619
Epoch 52/100
[1m46/46[0m [32m-----[0m[37m[0m [1m0s[0m
601ms/step - accuracy: 0.9523 - auc: 0.9964 - loss: 0.1308 -
precision: 0.9548 - recall: 0.9487

```

```

Epoch 52: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m[37m[0m [1m44s[0m
967ms/step - accuracy: 0.9738 - auc: 0.9988 - loss: 0.0853 -
precision: 0.9744 - recall: 0.9714 - val_accuracy: 0.8644 -
val_auc: 0.9810 - val_loss: 0.3570 - val_precision: 0.8706 -
val_recall: 0.8619
Epoch 52/100
[1m46/46[0m [32m-----[0m[37m[0m [1m0s[0m
601ms/step - accuracy: 0.9523 - auc: 0.9964 - loss: 0.1308 -
precision: 0.9548 - recall: 0.9487

```

```

Epoch 52: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m[37m[0m [1m44s[0m
967ms/step - accuracy: 0.9738 - auc: 0.9988 - loss: 0.0853 -
precision: 0.9744 - recall: 0.9714 - val_accuracy: 0.8644 -
val_auc: 0.9810 - val_loss: 0.3570 - val_precision: 0.8706 -
val_recall: 0.8619
Epoch 52/100
[1m46/46[0m [32m-----[0m[37m[0m [1m0s[0m
601ms/step - accuracy: 0.9523 - auc: 0.9964 - loss: 0.1308 -
precision: 0.9548 - recall: 0.9487

```

Epoch 52: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 52: finished saving model to

/kaggle/working/VGG19_checkpoint.keras

1m46/46[0m 32m-----[0m[37m[0m 1m44s[0m
966ms/step - accuracy: 0.9603 - auc: 0.9975 - loss: 0.1124 -
precision: 0.9617 - recall: 0.9579 - val_accuracy: 0.9306 -
val_auc: 0.9922 - val_loss: 0.1881 - val_precision: 0.9316 -
val_recall: 0.9281

Epoch 53/100

1m46/46[0m 32m-----[0m[37m[0m 1m0s[0m
601ms/step - accuracy: 0.9812 - auc: 0.9993 - loss: 0.0678 -
precision: 0.9825 - recall: 0.9788

Epoch 53: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 53: finished saving model to

/kaggle/working/VGG19_checkpoint.keras

1m46/46[0m 32m-----[0m[37m[0m 1m44s[0m
968ms/step - accuracy: 0.9836 - auc: 0.9992 - loss: 0.0637 -
precision: 0.9844 - recall: 0.9814 - val_accuracy: 0.9250 -
val_auc: 0.9907 - val_loss: 0.2126 - val_precision: 0.9293 -
val_recall: 0.9200

Epoch 54/100

1m46/46[0m 32m-----[0m[37m[0m 1m0s[0m
601ms/step - accuracy: 0.9854 - auc: 0.9995 - loss: 0.0557 -
precision: 0.9854 - recall: 0.9853

Epoch 54: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 54: finished saving model to

/kaggle/working/VGG19_checkpoint.keras

1m46/46[0m 32m-----[0m[37m[0m 1m44s[0m
968ms/step - accuracy: 0.9894 - auc: 0.9997 - loss: 0.0462 -
precision: 0.9896 - recall: 0.9891 - val_accuracy: 0.9388 -
val_auc: 0.9926 - val_loss: 0.1769 - val_precision: 0.9386 -
val_recall: 0.9362

Epoch 55/100

1m46/46[0m 32m-----[0m[37m[0m 1m0s[0m
603ms/step - accuracy: 0.9923 - auc: 0.9998 - loss: 0.0388 -
precision: 0.9923 - recall: 0.9919

Epoch 55: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 55: finished saving model to

/kaggle/working/VGG19_checkpoint.keras

1m46/46[0m 32m-----[0m[37m[0m 1m45s[0m
969ms/step - accuracy: 0.9887 - auc: 0.9996 - loss: 0.0478 -
precision: 0.9887 - recall: 0.9878 - val_accuracy: 0.8694 -
val_auc: 0.9734 - val_loss: 0.4231 - val_precision: 0.8729 -
val_recall: 0.8625

Epoch 56/100

1m46/46[0m 32m-----[0m[37m[0m 1m0s[0m
600ms/step - accuracy: 0.9021 - auc: 0.9885 - loss: 0.2468 -
precision: 0.9051 - recall: 0.8983

Epoch 56: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 56: finished saving model to

/kaggle/working/VGG19_checkpoint.keras

1m46/46[0m 32m-----[0m[37m[0m 1m44s[0m

966ms/step - accuracy: 0.9069 - auc: 0.9895 - loss: 0.2344 -
 precision: 0.9102 - recall: 0.9029 - val_accuracy: 0.9081 -
 val_auc: 0.9887 - val_loss: 0.2467 - val_precision: 0.9112 -
 val_recall: 0.9044

Epoch 57/100

1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m
 603ms/step - accuracy: 0.9782 - auc: 0.9993 - loss: 0.0645 -
 precision: 0.9805 - recall: 0.9777

Epoch 57: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 57: finished saving model to

/kaggle/working/VGG19_checkpoint.keras

1m46/46 [0m 32m] [0m 37m] [0m 1m45s] [0m
 970ms/step - accuracy: 0.9845 - auc: 0.9996 - loss: 0.0555 -
 precision: 0.9856 - recall: 0.9843 - val_accuracy: 0.9319 -
 val_auc: 0.9924 - val_loss: 0.1848 - val_precision: 0.9335 -
 val_recall: 0.9300

Epoch 58/100

1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m
 598ms/step - accuracy: 0.9932 - auc: 0.9998 - loss: 0.0385 -
 precision: 0.9939 - recall: 0.9932

Epoch 58: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 58: finished saving model to

/kaggle/working/VGG19_checkpoint.keras

1m46/46 [0m 32m] [0m 37m] [0m 1m44s] [0m
 964ms/step - accuracy: 0.9931 - auc: 0.9998 - loss: 0.0384 -
 precision: 0.9933 - recall: 0.9931 - val_accuracy: 0.9375 -
 val_auc: 0.9918 - val_loss: 0.1966 - val_precision: 0.9380 -
 val_recall: 0.9356

Epoch 59/100

1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m
 604ms/step - accuracy: 0.9857 - auc: 0.9997 - loss: 0.0472 -
 precision: 0.9860 - recall: 0.9854

Epoch 59: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 59: finished saving model to

/kaggle/working/VGG19_checkpoint.keras

1m46/46 [0m 32m] [0m 37m] [0m 1m45s] [0m
 971ms/step - accuracy: 0.9856 - auc: 0.9996 - loss: 0.0489 -
 precision: 0.9860 - recall: 0.9854 - val_accuracy: 0.9344 -
 val_auc: 0.9925 - val_loss: 0.1852 - val_precision: 0.9360 -
 val_recall: 0.9319

Epoch 60/100

1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m
 600ms/step - accuracy: 0.9954 - auc: 0.9999 - loss: 0.0301 -
 precision: 0.9956 - recall: 0.9947

Epoch 60: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 60: finished saving model to

/kaggle/working/VGG19_checkpoint.keras

1m46/46 [0m 32m] [0m 37m] [0m 1m44s] [0m
 968ms/step - accuracy: 0.9909 - auc: 0.9998 - loss: 0.0388 -
 precision: 0.9913 - recall: 0.9905 - val_accuracy: 0.9044 -
 val_auc: 0.9878 - val_loss: 0.2678 - val_precision: 0.9065 -
 val_recall: 0.9025

Epoch 61/100


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[1m46/46[0m [32m-----[0m[0m[37m[0m [1m0s[0m
602ms/step - accuracy: 0.9912 - auc: 0.9999 - loss: 0.0300 -
precision: 0.9914 - recall: 0.9912
Epoch 61: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 61: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m[0m[37m[0m [1m45s[0m
970ms/step - accuracy: 0.9927 - auc: 0.9999 - loss: 0.0312 -
precision: 0.9929 - recall: 0.9927 - val_accuracy: 0.9344 -
val_auc: 0.9935 - val_loss: 0.1708 - val_precision: 0.9355 -
val_recall: 0.9331
Epoch 62/100
[1m46/46[0m [32m-----[0m[0m[37m[0m [1m0s[0m
603ms/step - accuracy: 0.9925 - auc: 0.9999 - loss: 0.0334 -
precision: 0.9925 - recall: 0.9925
Epoch 62: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 62: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m[0m[37m[0m [1m45s[0m
972ms/step - accuracy: 0.9925 - auc: 0.9999 - loss: 0.0338 -
precision: 0.9925 - recall: 0.9925 - val_accuracy: 0.9431 -
val_auc: 0.9931 - val_loss: 0.1764 - val_precision: 0.9448 -
val_recall: 0.9413
Epoch 63/100
[1m46/46[0m [32m-----[0m[0m[37m[0m [1m0s[0m
599ms/step - accuracy: 0.9956 - auc: 1.0000 - loss: 0.0224 -
precision: 0.9956 - recall: 0.9955
Epoch 63: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 63: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m[0m[37m[0m [1m44s[0m
963ms/step - accuracy: 0.9960 - auc: 1.0000 - loss: 0.0231 -
precision: 0.9960 - recall: 0.9958 - val_accuracy: 0.9381 -
val_auc: 0.9924 - val_loss: 0.1803 - val_precision: 0.9398 -
val_recall: 0.9369
Epoch 64/100
[1m46/46[0m [32m-----[0m[0m[37m[0m [1m0s[0m
602ms/step - accuracy: 0.9983 - auc: 1.0000 - loss: 0.0205 -
precision: 0.9983 - recall: 0.9982
Epoch 64: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 64: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m[0m[37m[0m [1m44s[0m
968ms/step - accuracy: 0.9965 - auc: 1.0000 - loss: 0.0224 -
precision: 0.9965 - recall: 0.9962 - val_accuracy: 0.9375 -
val_auc: 0.9909 - val_loss: 0.2038 - val_precision: 0.9397 -
val_recall: 0.9344
Epoch 65/100
[1m46/46[0m [32m-----[0m[0m[37m[0m [1m0s[0m
604ms/step - accuracy: 0.9979 - auc: 1.0000 - loss: 0.0189 -
precision: 0.9979 - recall: 0.9979
Epoch 65: saving model to /kaggle/working/VGG19_checkpoint.keras

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Epoch 65: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m [37m[0m [1m45s[0m
969ms/step - accuracy: 0.9958 - auc: 1.0000 - loss: 0.0222 -
precision: 0.9958 - recall: 0.9958 - val_accuracy: 0.9469 -
val_auc: 0.9930 - val_loss: 0.1748 - val_precision: 0.9480 -
val_recall: 0.9456
Epoch 66/100
[1m46/46[0m [32m-----[0m [37m[0m [1m0s[0m
602ms/step - accuracy: 0.9984 - auc: 1.0000 - loss: 0.0165 -
precision: 0.9984 - recall: 0.9984
Epoch 66: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 66: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m [37m[0m [1m45s[0m
971ms/step - accuracy: 0.9987 - auc: 1.0000 - loss: 0.0146 -
precision: 0.9987 - recall: 0.9987 - val_accuracy: 0.9375 -
val_auc: 0.9925 - val_loss: 0.1765 - val_precision: 0.9398 -
val_recall: 0.9362
Epoch 67/100
[1m46/46[0m [32m-----[0m [37m[0m [1m0s[0m
601ms/step - accuracy: 0.9979 - auc: 1.0000 - loss: 0.0121 -
precision: 0.9979 - recall: 0.9979
Epoch 67: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 67: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m [37m[0m [1m44s[0m
968ms/step - accuracy: 0.9976 - auc: 1.0000 - loss: 0.0139 -
precision: 0.9976 - recall: 0.9976 - val_accuracy: 0.9444 -
val_auc: 0.9929 - val_loss: 0.1695 - val_precision: 0.9449 -
val_recall: 0.9431
Epoch 68/100
[1m46/46[0m [32m-----[0m [37m[0m [1m0s[0m
603ms/step - accuracy: 0.9982 - auc: 1.0000 - loss: 0.0123 -
precision: 0.9982 - recall: 0.9982
Epoch 68: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 68: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m [37m[0m [1m45s[0m
971ms/step - accuracy: 0.9980 - auc: 1.0000 - loss: 0.0132 -
precision: 0.9980 - recall: 0.9980 - val_accuracy: 0.9400 -
val_auc: 0.9926 - val_loss: 0.1906 - val_precision: 0.9411 -
val_recall: 0.9388
Epoch 69/100
[1m46/46[0m [32m-----[0m [37m[0m [1m0s[0m
600ms/step - accuracy: 0.9987 - auc: 1.0000 - loss: 0.0118 -
precision: 0.9987 - recall: 0.9987
Epoch 69: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 69: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m [37m[0m [1m44s[0m
966ms/step - accuracy: 0.9991 - auc: 1.0000 - loss: 0.0106 -
precision: 0.9991 - recall: 0.9991 - val_accuracy: 0.9444 -

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val_auc: 0.9930 - val_loss: 0.1711 - val_precision: 0.9449 -
val_recall: 0.9425

Epoch 70/100

1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m 601ms/step - accuracy: 0.9992 - auc: 1.0000 - loss: 0.0086 -
precision: 0.9992 - recall: 0.9992

Epoch 70: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 70: finished saving model to
/kaggle/working/VGG19_checkpoint.keras

1m46/46 [0m 32m] [0m 37m] [0m 1m45s] [0m 968ms/step - accuracy: 0.9978 - auc: 1.0000 - loss: 0.0111 -
precision: 0.9978 - recall: 0.9978 - val_accuracy: 0.9025 -
val_auc: 0.9883 - val_loss: 0.2798 - val_precision: 0.9059 -
val_recall: 0.9025

Epoch 71/100

1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m 602ms/step - accuracy: 0.9961 - auc: 1.0000 - loss: 0.0189 -
precision: 0.9961 - recall: 0.9960

Epoch 71: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 71: finished saving model to
/kaggle/working/VGG19_checkpoint.keras

1m46/46 [0m 32m] [0m 37m] [0m 1m45s] [0m 971ms/step - accuracy: 0.9976 - auc: 1.0000 - loss: 0.0148 -
precision: 0.9976 - recall: 0.9973 - val_accuracy: 0.9350 -
val_auc: 0.9913 - val_loss: 0.1994 - val_precision: 0.9360 -
val_recall: 0.9325

Epoch 72/100

1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m 599ms/step - accuracy: 0.9978 - auc: 1.0000 - loss: 0.0108 -
precision: 0.9978 - recall: 0.9978

Epoch 72: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 72: finished saving model to
/kaggle/working/VGG19_checkpoint.keras

1m46/46 [0m 32m] [0m 37m] [0m 1m44s] [0m 966ms/step - accuracy: 0.9976 - auc: 1.0000 - loss: 0.0136 -
precision: 0.9976 - recall: 0.9976 - val_accuracy: 0.8731 -
val_auc: 0.9688 - val_loss: 0.4948 - val_precision: 0.8763 -
val_recall: 0.8719

Epoch 73/100

1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m 604ms/step - accuracy: 0.9727 - auc: 0.9984 - loss: 0.0683 -
precision: 0.9729 - recall: 0.9727

Epoch 73: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 73: finished saving model to
/kaggle/working/VGG19_checkpoint.keras

1m46/46 [0m 32m] [0m 37m] [0m 1m44s] [0m 967ms/step - accuracy: 0.9594 - auc: 0.9960 - loss: 0.1164 -
precision: 0.9597 - recall: 0.9594 - val_accuracy: 0.9406 -
val_auc: 0.9901 - val_loss: 0.1993 - val_precision: 0.9430 -
val_recall: 0.9406

Epoch 74/100

1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m 601ms/step - accuracy: 0.9779 - auc: 0.9993 - loss: 0.0582 -

precision: 0.9781 - recall: 0.9778
 Epoch 74: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 74: finished saving model to
 /kaggle/working/VGG19_checkpoint.keras
 1m46/46[0m 32m-----[0m[37m[0m 1m44s[0m
 966ms/step - accuracy: 0.9774 - auc: 0.9994 - loss: 0.0564 -
 precision: 0.9780 - recall: 0.9772 - val_accuracy: 0.8725 -
 val_auc: 0.9751 - val_loss: 0.4286 - val_precision: 0.8728 -
 val_recall: 0.8706
 Epoch 75/100
 1m46/46[0m 32m-----[0m[37m[0m 1m0s[0m
 600ms/step - accuracy: 0.9895 - auc: 0.9997 - loss: 0.0338 -
 precision: 0.9904 - recall: 0.9885
 Epoch 75: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 75: finished saving model to
 /kaggle/working/VGG19_checkpoint.keras
 1m46/46[0m 32m-----[0m[37m[0m 1m45s[0m
 968ms/step - accuracy: 0.9953 - auc: 0.9999 - loss: 0.0200 -
 precision: 0.9956 - recall: 0.9951 - val_accuracy: 0.9381 -
 val_auc: 0.9926 - val_loss: 0.1820 - val_precision: 0.9398 -
 val_recall: 0.9362
 Epoch 76/100
 1m46/46[0m 32m-----[0m[37m[0m 1m0s[0m
 604ms/step - accuracy: 0.9732 - auc: 0.9984 - loss: 0.0710 -
 precision: 0.9740 - recall: 0.9728
 Epoch 76: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 76: finished saving model to
 /kaggle/working/VGG19_checkpoint.keras
 1m46/46[0m 32m-----[0m[37m[0m 1m44s[0m
 968ms/step - accuracy: 0.9814 - auc: 0.9989 - loss: 0.0541 -
 precision: 0.9818 - recall: 0.9812 - val_accuracy: 0.9425 -
 val_auc: 0.9921 - val_loss: 0.1834 - val_precision: 0.9425 -
 val_recall: 0.9419
 Epoch 77/100
 1m46/46[0m 32m-----[0m[37m[0m 1m0s[0m
 601ms/step - accuracy: 0.9989 - auc: 1.0000 - loss: 0.0117 -
 precision: 0.9989 - recall: 0.9989
 Epoch 77: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 77: finished saving model to
 /kaggle/working/VGG19_checkpoint.keras
 1m46/46[0m 32m-----[0m[37m[0m 1m44s[0m
 968ms/step - accuracy: 0.9991 - auc: 1.0000 - loss: 0.0103 -
 precision: 0.9991 - recall: 0.9991 - val_accuracy: 0.9431 -
 val_auc: 0.9927 - val_loss: 0.1840 - val_precision: 0.9435 -
 val_recall: 0.9400
 Epoch 78/100
 1m46/46[0m 32m-----[0m[37m[0m 1m0s[0m
 601ms/step - accuracy: 0.9995 - auc: 1.0000 - loss: 0.0084 -
 precision: 0.9995 - recall: 0.9995
 Epoch 78: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 78: finished saving model to
 /kaggle/working/VGG19_checkpoint.keras

```

[1m46/46][0m [32m-----[0m[37m[0m [1m44s[0m
966ms/step - accuracy: 0.9987 - auc: 1.0000 - loss: 0.0097 -
precision: 0.9987 - recall: 0.9987 - val_accuracy: 0.9438 -
val_auc: 0.9929 - val_loss: 0.1750 - val_precision: 0.9443 -
val_recall: 0.9438

```

Epoch 79/100

```

[1m46/46][0m [32m-----[0m[37m[0m [1m0s[0m
602ms/step - accuracy: 0.9898 - auc: 0.9999 - loss: 0.0267 -
precision: 0.9901 - recall: 0.9891

```

Epoch 79: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 79: finished saving model to
/kaggle/working/VGG19_checkpoint.keras

```

[1m46/46][0m [32m-----[0m[37m[0m [1m45s[0m
970ms/step - accuracy: 0.9936 - auc: 1.0000 - loss: 0.0196 -
precision: 0.9938 - recall: 0.9934 - val_accuracy: 0.9400 -
val_auc: 0.9931 - val_loss: 0.1818 - val_precision: 0.9418 -
val_recall: 0.9400

```

Epoch 80/100

```

[1m46/46][0m [32m-----[0m[37m[0m [1m0s[0m
598ms/step - accuracy: 0.9997 - auc: 1.0000 - loss: 0.0061 -
precision: 0.9997 - recall: 0.9997

```

Epoch 80: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 80: finished saving model to
/kaggle/working/VGG19_checkpoint.keras

```

[1m46/46][0m [32m-----[0m[37m[0m [1m44s[0m
962ms/step - accuracy: 0.9996 - auc: 1.0000 - loss: 0.0063 -
precision: 0.9996 - recall: 0.9996 - val_accuracy: 0.9431 -
val_auc: 0.9927 - val_loss: 0.1804 - val_precision: 0.9437 -
val_recall: 0.9425

```

Epoch 81/100

```

[1m46/46][0m [32m-----[0m[37m[0m [1m0s[0m
604ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0055 -
precision: 1.0000 - recall: 1.0000

```

Epoch 81: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 81: finished saving model to
/kaggle/working/VGG19_checkpoint.keras

```

[1m46/46][0m [32m-----[0m[37m[0m [1m45s[0m
972ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0050 -
precision: 1.0000 - recall: 1.0000 - val_accuracy: 0.9506 -
val_auc: 0.9927 - val_loss: 0.1737 - val_precision: 0.9518 -
val_recall: 0.9506

```

Epoch 82/100

```

[1m46/46][0m [32m-----[0m[37m[0m [1m0s[0m
599ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0045 -
precision: 1.0000 - recall: 1.0000

```

Epoch 82: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 82: finished saving model to
/kaggle/working/VGG19_checkpoint.keras

```

[1m46/46][0m [32m-----[0m[37m[0m [1m44s[0m
965ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0044 -
precision: 1.0000 - recall: 1.0000 - val_accuracy: 0.9475 -
val_auc: 0.9924 - val_loss: 0.1819 - val_precision: 0.9480 -
val_recall: 0.9463

```

Epoch 83/100

1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m 601ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0046 - precision: 1.0000 - recall: 1.0000
Epoch 83: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 83: finished saving model to
/kaggle/working/VGG19_checkpoint.keras

1m46/46 [0m 32m] [0m 37m] [0m 1m44s] [0m 965ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0041 - precision: 1.0000 - recall: 1.0000 - val_accuracy: 0.9469 - val_auc: 0.9923 - val_loss: 0.1790 - val_precision: 0.9475 - val_recall: 0.9469

Epoch 84/100

1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m 601ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0036 - precision: 1.0000 - recall: 1.0000
Epoch 84: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 84: finished saving model to
/kaggle/working/VGG19_checkpoint.keras

1m46/46 [0m 32m] [0m 37m] [0m 1m44s] [0m 966ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0035 - precision: 1.0000 - recall: 1.0000 - val_accuracy: 0.9500 - val_auc: 0.9925 - val_loss: 0.1734 - val_precision: 0.9499 - val_recall: 0.9488

Epoch 85/100

1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m 599ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0034 - precision: 1.0000 - recall: 1.0000
Epoch 85: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 85: finished saving model to
/kaggle/working/VGG19_checkpoint.keras

1m46/46 [0m 32m] [0m 37m] [0m 1m44s] [0m 964ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0040 - precision: 1.0000 - recall: 1.0000 - val_accuracy: 0.9513 - val_auc: 0.9924 - val_loss: 0.1737 - val_precision: 0.9518 - val_recall: 0.9500

Epoch 86/100

1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m 601ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0037 - precision: 1.0000 - recall: 1.0000
Epoch 86: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 86: finished saving model to
/kaggle/working/VGG19_checkpoint.keras

1m46/46 [0m 32m] [0m 37m] [0m 1m45s] [0m 971ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0035 - precision: 1.0000 - recall: 1.0000 - val_accuracy: 0.9488 - val_auc: 0.9931 - val_loss: 0.1788 - val_precision: 0.9492 - val_recall: 0.9463

Epoch 87/100

1m46/46 [0m 32m] [0m 37m] [0m 1m0s] [0m 600ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0034 - precision: 1.0000 - recall: 1.0000
Epoch 87: saving model to /kaggle/working/VGG19_checkpoint.keras

```

Epoch 87: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m [37m[0m [1m45s[0m
969ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0040 -
precision: 1.0000 - recall: 1.0000 - val_accuracy: 0.9500 -
val_auc: 0.9928 - val_loss: 0.1782 - val_precision: 0.9500 -
val_recall: 0.9494
Epoch 88/100
[1m46/46[0m [32m-----[0m [37m[0m [1m0s[0m
601ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0037 -
precision: 1.0000 - recall: 1.0000
Epoch 88: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 88: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m [37m[0m [1m45s[0m
969ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0033 -
precision: 1.0000 - recall: 1.0000 - val_accuracy: 0.9519 -
val_auc: 0.9908 - val_loss: 0.1855 - val_precision: 0.9524 -
val_recall: 0.9506
Epoch 89/100
[1m46/46[0m [32m-----[0m [37m[0m [1m0s[0m
601ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0025 -
precision: 1.0000 - recall: 1.0000
Epoch 89: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 89: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m [37m[0m [1m45s[0m
969ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0026 -
precision: 1.0000 - recall: 1.0000 - val_accuracy: 0.9481 -
val_auc: 0.9923 - val_loss: 0.1852 - val_precision: 0.9499 -
val_recall: 0.9475
Epoch 90/100
[1m46/46[0m [32m-----[0m [37m[0m [1m0s[0m
601ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0026 -
precision: 1.0000 - recall: 1.0000
Epoch 90: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 90: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m [37m[0m [1m45s[0m
969ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0024 -
precision: 1.0000 - recall: 1.0000 - val_accuracy: 0.9500 -
val_auc: 0.9925 - val_loss: 0.1798 - val_precision: 0.9506 -
val_recall: 0.9500
Epoch 91/100
[1m46/46[0m [32m-----[0m [37m[0m [1m0s[0m
601ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0024 -
precision: 1.0000 - recall: 1.0000
Epoch 91: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 91: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
[1m46/46[0m [32m-----[0m [37m[0m [1m44s[0m
967ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0023 -

```

```
Epoch 92: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
1m46/460m 32m-----1m45s0m
971ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0023 -
precision: 1.0000 - recall: 1.0000 - val_accuracy: 0.9413 -
val_auc: 0.9921 - val_loss: 0.1957 - val_precision: 0.9418 -
val_recall: 0.9406
Epoch 93/100
1m46/460m 32m-----1m45s0m
599ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0022 -
precision: 1.0000 - recall: 1.0000
Epoch 93: saving model to /kaggle/working/VGG19_checkpoint.keras
```

```
Epoch 94: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
1m46/460m 32m-----1m45s0m
973ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0018 -
precision: 1.0000 - recall: 1.0000 - val_accuracy: 0.9519 -
val_auc: 0.9918 - val_loss: 0.1805 - val_precision: 0.9530 -
val_recall: 0.9513
Epoch 95/100
1m46/460m 32m-----1m40s0m
602ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0017 -
precision: 1.0000 - recall: 1.0000
Epoch 95: saving model to /kaggle/working/VGG19_checkpoint.keras
```

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602ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0016 -
precision: 1.0000 - recall: 1.0000
Epoch 96: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 96: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
1m46/46[0m 32m-----[0m[37m[0m 1m45s[0m
979ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0019 -
precision: 1.0000 - recall: 1.0000 - val_accuracy: 0.9362 -
val_auc: 0.9913 - val_loss: 0.2139 - val_precision: 0.9374 -
val_recall: 0.9362

Epoch 97/100
1m46/46[0m 32m-----[0m[37m[0m 1m0s[0m
604ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0021 -
precision: 1.0000 - recall: 1.0000
Epoch 97: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 97: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
1m46/46[0m 32m-----[0m[37m[0m 1m45s[0m
981ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0020 -
precision: 1.0000 - recall: 1.0000 - val_accuracy: 0.9481 -
val_auc: 0.9919 - val_loss: 0.1812 - val_precision: 0.9481 -
val_recall: 0.9475

Epoch 98/100
1m46/46[0m 32m-----[0m[37m[0m 1m0s[0m
603ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0017 -
precision: 1.0000 - recall: 1.0000
Epoch 98: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 98: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
1m46/46[0m 32m-----[0m[37m[0m 1m45s[0m
974ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0016 -
precision: 1.0000 - recall: 1.0000 - val_accuracy: 0.9500 -
val_auc: 0.9906 - val_loss: 0.1926 - val_precision: 0.9512 -
val_recall: 0.9494

Epoch 99/100
1m46/46[0m 32m-----[0m[37m[0m 1m0s[0m
602ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0014 -
precision: 1.0000 - recall: 1.0000
Epoch 99: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 99: finished saving model to
/kaggle/working/VGG19_checkpoint.keras
1m46/46[0m 32m-----[0m[37m[0m 1m45s[0m
982ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0013 -
precision: 1.0000 - recall: 1.0000 - val_accuracy: 0.9500 -
val_auc: 0.9904 - val_loss: 0.1884 - val_precision: 0.9512 -
val_recall: 0.9500

Epoch 100/100
1m46/46[0m 32m-----[0m[37m[0m 1m0s[0m
608ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0013 -
precision: 1.0000 - recall: 1.0000
Epoch 100: saving model to /kaggle/working/VGG19_checkpoint.keras

Epoch 100: finished saving model to

```

/kaggle/working/VGG19_checkpoint.keras
[1m46/46][0m  [32m-----[0m[37m[0m  [1m46s[0m
991ms/step - accuracy: 1.0000 - auc: 1.0000 - loss: 0.0013 -
precision: 1.0000 - recall: 1.0000 - val_accuracy: 0.9475 -
val_auc: 0.9911 - val_loss: 0.1897 - val_precision: 0.9481 -
val_recall: 0.9469
[1m10/10][0m  [32m-----[0m[37m[0m  [1m22s[0m
2s/step - accuracy: 0.9495 - auc: 0.9871 - loss: 0.2133 -
precision: 0.9495 - recall: 0.9495

```

Test Results:

Loss: 0.2133, Accuracy: 0.9495, Precision: 0.9495, Recall: 0.9495,
AUC: 0.9871

```
In [5]: model.save("VGG19_D1.h5")
        model.save("VGG19_D1.keras")
```

WARNING:absl:You are saving your model as an HDF5 file via
`model.save()` or `keras.saving.save\_model(model)`. This file
format is considered legacy. We recommend using instead the
native Keras format, e.g. `model.save('my\_model.keras')` or
`keras.saving.save\_model(model, 'my\_model.keras')`.

```
In [6]: model.summary()
```

Model: "sequential_1"

Layer (type)	Output Shape	
vgg19 (Functional)	(None, 7, 7, 512)	20,
flatten_1 (Flatten)	(None, 25088)	
dense_3 (Dense)	(None, 4096)	102,
dense_4 (Dense)	(None, 4096)	16,
dense_5 (Dense)	(None, 4)	



Total params: 378,711,118 (1.41 GB)

Trainable params: 119,562,244 (456.09 MB)

Non-trainable params: 20,024,384 (76.39 MB)

Optimizer params: 239,124,490 (912.19 MB)

```
In [7]: with open(history_path, "rb") as f:
        history = pickle.load(f)

        print("Available keys in history:", history.keys())

        # Accuracy & Loss
        plt.figure(figsize=(12, 6))
        plt.subplot(1, 2, 1)
        plt.plot(history['accuracy'], label='Train Accuracy')
        plt.plot(history['val_accuracy'], label='Validation Accuracy')
        plt.title('Model Accuracy'); plt.xlabel('Epoch');
        plt.ylabel('Accuracy'); plt.legend()

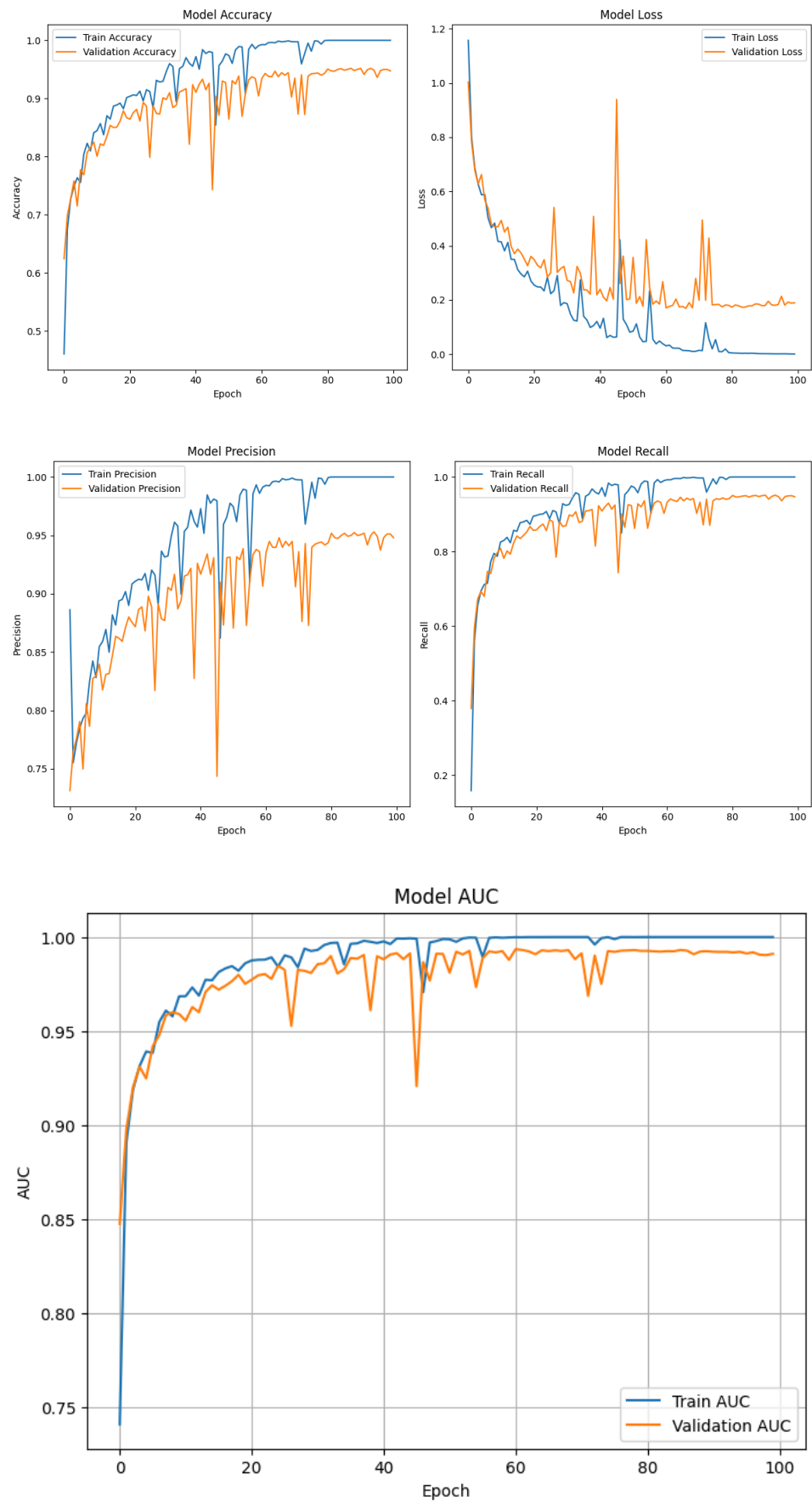
        plt.subplot(1, 2, 2)
        plt.plot(history['loss'], label='Train Loss')
        plt.plot(history['val_loss'], label='Validation Loss')
        plt.title('Model Loss'); plt.xlabel('Epoch'); plt.ylabel('Loss');
        plt.legend()
        plt.tight_layout()
        plt.show()

        # Precision & Recall
        plt.figure(figsize=(12, 6))
        plt.subplot(1, 2, 1)
        plt.plot(history['precision'], label='Train Precision')
        plt.plot(history['val_precision'], label='Validation Precision')
        plt.title('Model Precision'); plt.xlabel('Epoch');
        plt.ylabel('Precision'); plt.legend()

        plt.subplot(1, 2, 2)
        plt.plot(history['recall'], label='Train Recall')
        plt.plot(history['val_recall'], label='Validation Recall')
        plt.title('Model Recall'); plt.xlabel('Epoch'); plt.ylabel('Recall');
        plt.legend()
        plt.tight_layout()
        plt.show()

        # AUC
        plt.figure(figsize=(8, 6))
        plt.plot(history['auc'], label='Train AUC')
        plt.plot(history['val_auc'], label='Validation AUC')
        plt.title('Model AUC'); plt.xlabel('Epoch'); plt.ylabel('AUC');
        plt.legend()
        plt.grid(True)
        plt.show()
```

```
Available keys in history: dict_keys(['accuracy', 'val_accuracy',
    'loss', 'val_loss', 'precision', 'val_precision', 'recall',
    'val_recall', 'auc', 'val_auc'])
```



```
In [8]: test_generator.reset()
predictions = model.predict(test_generator, verbose=1)
predicted_classes = np.argmax(predictions, axis=1)
true_labels = test_generator.classes

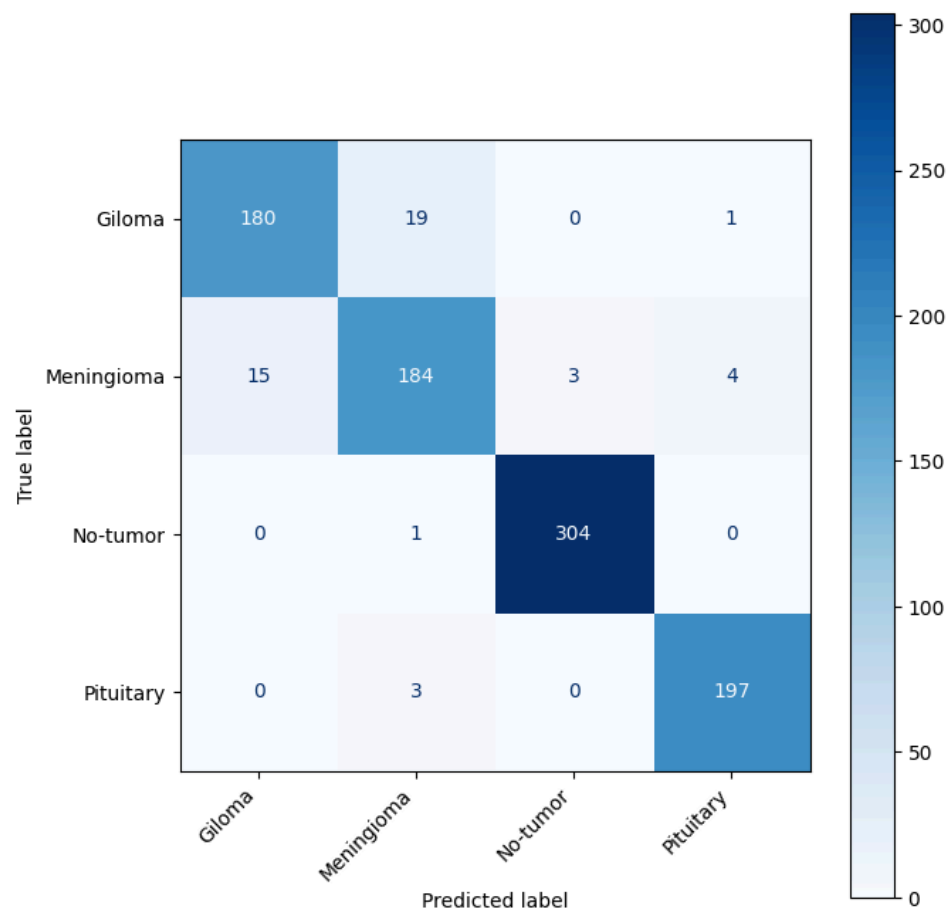
print("Class labels:", class_labels)
```

2[1m10/102[0m 2[32m—————2[0m2[37m2[0m 2[1m7s2[0m
622ms/step
Class labels: ['Glioma', 'Meningioma', 'No-tumor', 'Pituitary']

```
In [9]: from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

cm = confusion_matrix(true_labels, predicted_classes)
disp = ConfusionMatrixDisplay(confusion_matrix=cm,
display_labels=class_labels)

fig, ax = plt.subplots(figsize=(7, 7))
disp.plot(cmap=plt.cm.Blues, ax=ax)
plt.xticks(rotation=45, ha='right', fontsize=10)
plt.yticks(fontsize=10)
plt.tight_layout()
plt.show()
```



```

In [10]: def fetch_all_images(generator):
    images, labels = [], []
    for _ in range(len(generator)):
        batch_images, batch_labels = next(generator)
        images.extend(batch_images)
        labels.extend(batch_labels)
    return np.array(images), np.array(labels)

test_images, test_labels = fetch_all_images(test_generator)
predictions = model.predict(test_images)
predicted_classes = np.argmax(predictions, axis=1)
true_classes = np.argmax(test_labels, axis=1)

def plot_images(images, true_labels, predicted_classes, class_labels,
num_images=5, cols=5):
    num_classes = len(class_labels)
    num_per_class = num_images // num_classes
    selected_images, selected_true_labels, selected_predicted_classes =
    [], [], []

    for class_index in range(num_classes):
        indices = np.where(true_labels == class_index)[0]
        if len(indices) > 0:
            num_samples = min(num_per_class, len(indices))
            selected_indices = np.random.choice(indices, num_samples,
replace=False)
            selected_images.extend(images[selected_indices])
            selected_true_labels.extend(true_labels[selected_indices])

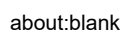
        selected_predicted_classes.extend(predicted_classes[selected_indices])
        else:
            print(f"No images available for class
{class_labels[class_index]}")

    num_images = len(selected_images)
    rows = (num_images + cols - 1) // cols
    plt.figure(figsize=(15, rows * 3))
    for i in range(num_images):
        plt.subplot(rows, cols, i + 1)
        plt.imshow(selected_images[i])
        plt.axis('off')
        plt.title(f"True: {class_labels[selected_true_labels[i]]}\nPred:
{class_labels[selected_predicted_classes[i]]}")
    plt.tight_layout()
    plt.show()

plot_images(test_images, true_classes, predicted_classes, class_labels,
num_images=batch_size, cols=5)

```

1m29/29 0m 32m ————— 0m 37m 0m 1m30s 0m
513ms/step





```
In [11]: from tensorflow.keras.utils import plot_model
from IPython.display import Image

plot_model(model, to_file='convnet.png', show_shapes=True,
show_layer_names=True)
Image(filename='convnet.png')
```

Out[11]:

